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SiViS: Simulated multi-patient physiological clinical states for advanced vital sign radar monitoring research

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Continuous, clinically reliable non-contact monitoring of multiple patients' vital signs remains a significant technological challenge in care settings. This paper introduces a systematically constructed radar-based dataset designed to test methods for simultaneous multi-patient vital sign monitoring. Developed with Ostrava University Training Hospital, the dataset includes recordings from two-three advanced medical simulation mannequins (SimMan 3G Plus), emulating a wide range of physiological states—from healthy resting to acute emergencies (apnea, cardiac arrest, severe respiratory distress). We varied sensor geometry (top, frontal, lateral views at 1-4 m, 0°/45° angles) and radar parameters (ADC samples, chirp loops, ramp times, frame rates, gains), yielding over 100 uniquely configured sessions. Preliminary beamforming-based processing achieves mean heart-rate and breathing-rate errors near clinical thresholds (MAE $\approx$ 6.6 bpm for HR, 1.47 bpm for RR), demonstrating the dataset's utility for developing advanced signal processing and machine-learning pipelines. In keeping with FAIR principles, all data are fully documented and publicly accessible, supporting reproducible research toward noninvasive multi-patient monitoring systems.

Background & Summary

Accurate and continuous monitoring of vital signs, such as heart rate (HR) and respiratory rate (RR), is essential for evaluating physiological function, diagnosing disease, and tracking patient recovery. In clinical practice, conventional monitoring techniques, such as electrocardiography or overnight polysomnography, require physical contact via electrodes or wearable sensors. These methods can be intrusive, costly, and uncomfortable, thus affecting patient compliance and potentially biasing measurements[1,2].

The need for scalable, comfortable, and long-term monitoring solutions has led to the growing interest in noncontact radio frequency (RF) based systems, particularly Doppler and frequency-modulated continuous-wave (FMCW) radars. These systems offer the ability to detect micro-movements of the chest associated with respiration and heartbeat without direct skin contact, making them attractive for in-home, nursing homes and clinical applications. However, current systems are predominantly optimized for single-person scenarios and often rely on bulky antenna arrays or complex signal processing pipelines that are not suited for practical multi-patient environments[3,4]. Simultaneous monitoring of multiple individuals is increasingly critical in real-world settings such as multi-bed hospital rooms, nursing homes, sleep laboratories, and home healthcare environments. In such scenarios, clinicians must assess the physiological states of several patients concurrently and without physical disturbance. multi-patient radar-based systems provide a compelling solution by offering real-time, nonintrusive monitoring, while eliminating the logistical constraints of wearable devices[2,5].

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The challenges associated with this task are complex. Recent advancements have demonstrated accurate extraction of HR, RR, and other physiological parameters in controlled laboratory conditions using both commercial and custom sensor systems. However, these efforts predominantly focus on single-subject Set-ups and short duration recordings, thereby failing to capture the complexity of real-world clinical or in-home multi-patient environments[6,7]. Multipath propagation, inter-subject interference, and limited spatial resolution complicate the separation and identification of individual vital sign signals. To address these, advanced algorithms leveraging sparsity, self-injection locking (SIL), and dictionary-based decomposition have been proposed, demonstrating improved robustness and accuracy in cluttered environments[1,3]. Despite the technical feasibility demonstrated in literature, several critical limitations persist. Existing datasets are generally small in scale, and lack health conditions diversity,[8,9]. Furthermore, multi-patient datasets are scarce, and when available, they often lack synchronized ground truth from gold-standard devices, limiting their utility for benchmarking advanced algorithms.

Using mmWave radars has shown potential for noninvasive monitoring of multiple individuals. However, challenges remain in reliably separating overlapping physiological signals, mitigating multipath interference, and ensuring robustness in cluttered indoor environments[3,10]. The majority of radar datasets either simulate multi-patient conditions or include limited real-world interactions, providing insufficient variability for training and testing deep learning or sparsity-based signal separation models[8,11]. Moreover, existing databases rarely integrate sensor parameters diversity in a single synchronized dataset. This restricts opportunities for sensor application research and algorithm development[8]. The lack of long-duration, multi-patient recordings, including variable patient health conditions, and radar location and settings, creates a substantial gap between current research and clinical deployment[12,13].

Unlike prior datasets that assume subjects are spatially separated, SiViS intentionally includes multi-patient scenes with overlapping range-azimuth positions, representing clinically realistic challenges where standard RA-based separation fails.

Objective. This work introduces an open-access radar signal dataset tailored for multi-patient, noncontact vital sign monitoring in simulated clinical environments. It addresses key limitations of existing datasets by providing long-duration recordings using advanced medical mannequins under controlled but clinically realistic scenarios that reflect physiological variability.

The dataset includes synchronized radar measurements from multiple spatial perspectives (top, frontal, lateral), distances (1-4 m), and orientation angles (0°, 45°). Each session simulates a variety of vital sign conditions, including abnormal respiratory and heart rates, enabling robust evaluation of signal processing and machine learning algorithms in various clinical states. Its difference with other radar vital sign datasets is that it includes variations in radar chirp configurations to study signal interference, multipath effects, and inter-subject variability in multi-patient settings. This dataset was designed to support reproducible research, the dataset adheres to FAIR principles (Findable, Accessible, Interoperable, Reusable)

The collaboration with Ostrava University Training Hospital ensured clinical relevance. Their simulation infrastructure and medical expertise enabled the creation of realistic emergency scenarios and multi-patient room Set-ups using high-fidelity mannequins. This partnership grounded the data acquisition protocol in real-world medical practices, improving the translational value of the data set for the development and implementation of algorithms in healthcare applications.

Clinical Relevance. The integration of radar technology in clinical respiratory monitoring represents a revolutionary advancement, addressing critical healthcare needs through accurate vital sign assessment. Respiratory abnormalities such as bradypnea, tachypnea, and apnea are indicative of severe clinical conditions, including acute coronary syndrome (ACS) and respiratory distress; thus, timely and precise monitoring are crucial for effective medical interventions[14,15]. However, there are not radar based vital sign datasets that are clinical validated that present this clinical conditions. Radar-based methods enables detection of subtle variations in breathing patterns without the discomfort or limitations inherent in traditional contact-based methods, offering significant improvements in patient compliance and long-term monitoring capabilities[16]. This technological approach is particularly relevant given the high incidence and clinical complexity associated with in-hospital cardiac arrests (IHCAs), where early detection significantly impacts patient survival[17,18]. SiVis database includes diverse simulated scenarios ranging of HR from 50 to 160 bpm and RR from 0-33. This variations in dataset provides a realistic and ethically training environment through the use of advanced medical mannequins in multi-patient conditions. This dataset will significantly improve algorithm development and training protocols and, will ensure more reliability for end-user solutions[19,20].

Limitations. Although mannequin-based simulations allow precise and ethical control of physiological parameters, they cannot fully replicate micro-motion variability and bio-mechanical coupling found in real patients. Future datasets will integrate hybrid acquisitions combining simulated and human data.

Methods

Radars. Radar data was acquired using the millimeter-wave mm-wave FMCW Texas Instruments (TI) IWR6843AOP radar, integrated to a DCA1000EVM board for high-speed ADC data capture. TI mmWave Studio software was used to configure the radar and DCA1000 module, as well as for data acquisition. The setup also required additional software dependencies, including the MATLAB Runtime Engine (version 8.5.1 or later) and Code Composer Studio (version 7.1 or later). This radar has a 77 to 81 GHz band and antenna configuration for 3TX and 4 TX. A second radar sensor was used as reference and comparison of radar technologies in the "Area

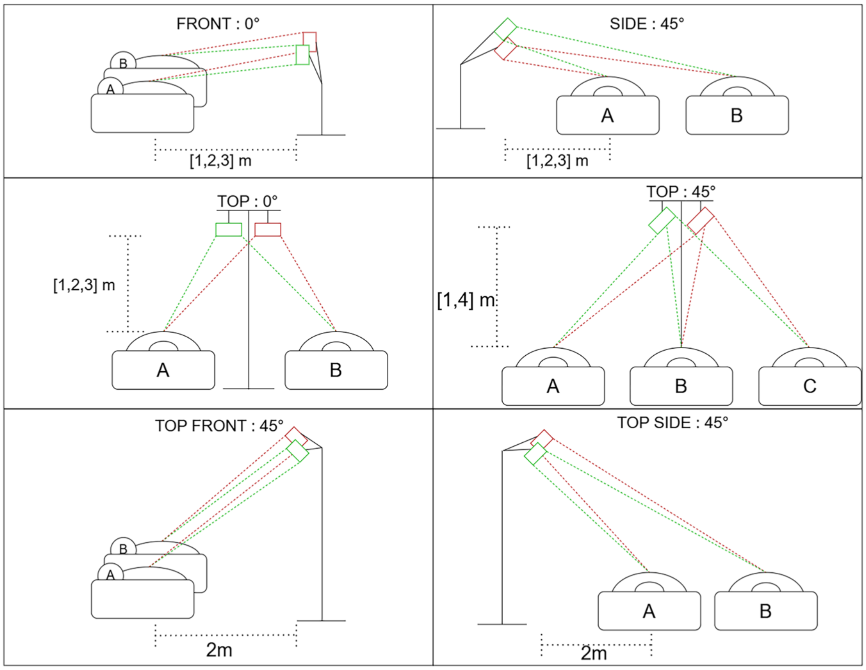

Fig. 1 Radar and Mannequin Area Set Up.

Set Up" measurements and the "Mannequin Set Up" measurements, the Infineon XENSIVTM BGT60LTR11AIP, a low-power 60 GHz Doppler radar sensor, and the recordings were done using the Radar Fusion GUI v3.6.5.

Physiological Scenarios and Mannequin Simulation. Three full-body patient simulator *SimMan® 3G PLUS (Laerdal Medical)* were used to generate controlled respiratory and cardiac activity. Only simulated respiration and heart rate characteristics were used. Respiratory simulation included spontaneous breathing with adjustable rate and depth and bilateral chest movement. Cardiac activity was generated using programmable heart rates from the internal ECG rhythm library, recorded from palpable blood pressure movements in the neck. All vital signs were controlled through the LLEAP software interface. No additional physiological features or clinical interventions were used. *The SimMan® 3G PLUS* mannequin allows digital programming and internal monitoring of vital signs via the interface, ensuring that heart rate and respiratory rate values are precisely defined and time-synchronized with radar acquisitions. These programmed parameters constitute the dataset's reference ground truth[21].

The dataset includes recordings from the following clinical scenarios:

- **A. Healthy resting state**: Normal respiration rate (RR = 14) and heart rate (HR = 80).
- **B. Healthy sleeping**: Reduced RR (12) and HR (60), simulating a relaxed state.
- **C. Healthy agitated**: Elevated RR (20) and HR (110), simulating mild stress or physical activity.
- **D. Apnea**: No respiratory activity (RR = 0), HR maintained at 80.
- **E. Asthma**: Increased RR (22) and moderate HR (90), simulating mild respiratory obstruction.
- **F. Asthma attack**: Severe respiratory distress (RR = 33), elevated HR (130).
- **G. Pneumothorax**: High RR (30) and HR (120), simulating compromised lung function.
- **H. Bradypnea**: Reduced breathing rate (RR = 6), low HR (50).
- **I. Cardiac arrest**: High HR (160) with RR maintained at 16.
- **J. Acute Coronary Syndrome (ACS) / Chest pain**: RR = 18, HR = 104.
- **K. Tachypnea / Hyperventilation**: Rapid breathing (RR = 30) with high HR (125).
- **L. Irregular breathing**: Manually varied RR (10, 6, 14, 1 minute each), constant HR = 110.

Each scenario was reproduced with fixed simulator parameters to ensure repeatability for signal processing and algorithm validation. Scenarios include multi-patient recordings (two or three mannequins), recorded in the emergency room of the training hospital, emulating typical hospital or home care room configurations. Each session was recorded as shown in the following images:

Acquisition Configurations. The recordings were collected in varying geometric and system-level configurations as shown in Fig. 1, as well as Chirp configurations to assess signal robustness and algorithmic generalization. Figure 2A shows an image of the 3 mannequin measurement with top view at 3 meters configured for the "Radar Set Up" measurements, while Fig. 2B shows an image of the 2 mannequin measurement with top view at 2 meters configured for the "Radar Set Up" measurements. The acquisition were done over three variable scenarios, "Area Set Up", where two mannequins were measured in all recordings, and the variables for measurement are

Fig. 2 Comparison of top views at different heights and number of mannequins (**A**) Top view at 4 m with 3 mannequins (**B**) TopSide view at 2 m with 2 mannequins.

based on: **Radar positions**: Top, Frontal and Lateral views, **Distances**: 1.0 m, 2.0 m, 3.0 m and 4.0 m and, **Radar Orientation angles**: 0°, 45°.

For the "Mannequin Set Up", there was a fixed "TopFront" position of the radar, and there are measurements with two mannequins at two meters height and three mannequins at four meters height. The variables were conditions A-L set randomly and aleatory to mannequins.

For this two Set Ups, standard radar configuration in the IWR68430AOP was set to: 16-bit ADC with normalized complex interleaved data format, 256 ADC samples, 128 Chirp loops, 3 Tx 4 Rx, Frame Periodicity 40, ADC Sample rate = 1000 Msps, Ramp End time = 60 μs, Receiver Gain 30 dB, Frequency Slope 30 MHz/μs. Also, standard radar configuration of the BGT60LTR11AIP was set to: 61.044 GHz, 12-bit ADC with normalized complex interleaved data format, sampling rate of 2 kHz, 256 ADC samples per chirp, 512 samples per frame, 1 TX 1 RX, Pulse repetition 500 μs, and each frame spans 128 ms.

For the "Radar Set Up", the Chirp parameters were settled only on IWR68430AOP, including variation in: **ADC sample count:** 256, 128, 64, **Chirps per loop:** 128, 64, 32, **Frame periodicity:** 60 ms, 40 ms, 24 ms, **ADC sampling rate:** 10 Msps, **Ramp end time:** 80 μs, 60 μs, **Receiver gain:** 40 dB, 30 dB, item[**Frequency slope:** 60 MHz/μs, 30 MHz/μs.

Data Records

The dataset is deposited in the Harvard Dataverse repository as SiViS - Simulated multi-patient physiological clinical states[22], available at https://doi.org/10.7910/DVN/S9SNAK.

Data Format. The raw radar data includes two formats :

I. From IWR6843AOP:

A. The Complex samples across all receiver channels (4RX) are stored in the .bin format. "The saved file has noninterleaved format beginning with the real part of every two samples and followed by the imaginary part of the every two samples. If there is a total of M number of chirps, the data is stored beginning with first chirp and ending with the Mth chirp. As the maximum data transfer capacity of DCA allows approx 1 GB per file, for some recordings there are 1-3 files ending in 0.bin, 1.bin, and 2.bin.

B. In order to follow FAIR principles, all files were converted to csv format. The acquired raw binary data from IWR6843 was decoded using a custom algorithm (Algorithm 1) based on TI DCA1000 Data Format instructions[23]. The binary is read as 16-bit signed integers, interleaved into in-phase (I) and quadrature (Q) components, and then reshaped into a complex-valued array. The data is segmented by frames and by receiver antenna, columns 1-256 belong to RX1, 257-512 to RX2, etc. Also, in this processing, concatenation has been done to store just 1 csv file per measurement. This structure is ready for further signal processing and analysis of the individual antenna input.

II. For the BGT60LTR11AIP:

A. A txt file is generated. The file starts with a header that contains metadata and parameter definitions. Each data segment is preceded by per-frame metadata (starting with #), After each per-frame header, the actual radar data values are stored as sequence of floating-point values, one per line. The data is written in pairs per sample: Each complex sample consists of two consecutive values: [I0, Q0, I1, Q1, …, IN, QN] where N = samples per frame - 1.

DataBase Structure. The SiVis dataset folder is organized into the three main experimental Set-ups: Area, Mannequin, and Radar Set-up. These divisions were designed to facilitate the development and benchmarking of algorithms to specific research objectives.

The Area Set-up enables systematic testing of optimal radar placement, angle, and position within a given space, as supporting studies on spatial configuration for improved signal acquisition. The Mannequin Set-up introduces the range of clinical conditions using two or three mannequins simultaneously, providing diverse training scenarios for the development of robust respiration and heart rate detection models, also allowing the comparison of two radar technologies. Finally, the Radar Set-up explores variations in radar acquisition parameters, offering, for the first time in vital sign radar datasets, an opportunity to systematically evaluate the impact of different chirp configurations and identify the optimal signal processing conditions for physiological monitoring of multi-patient challenges. Those data bases were structured as shown in Fig. 3. Each file contains the following encoding:

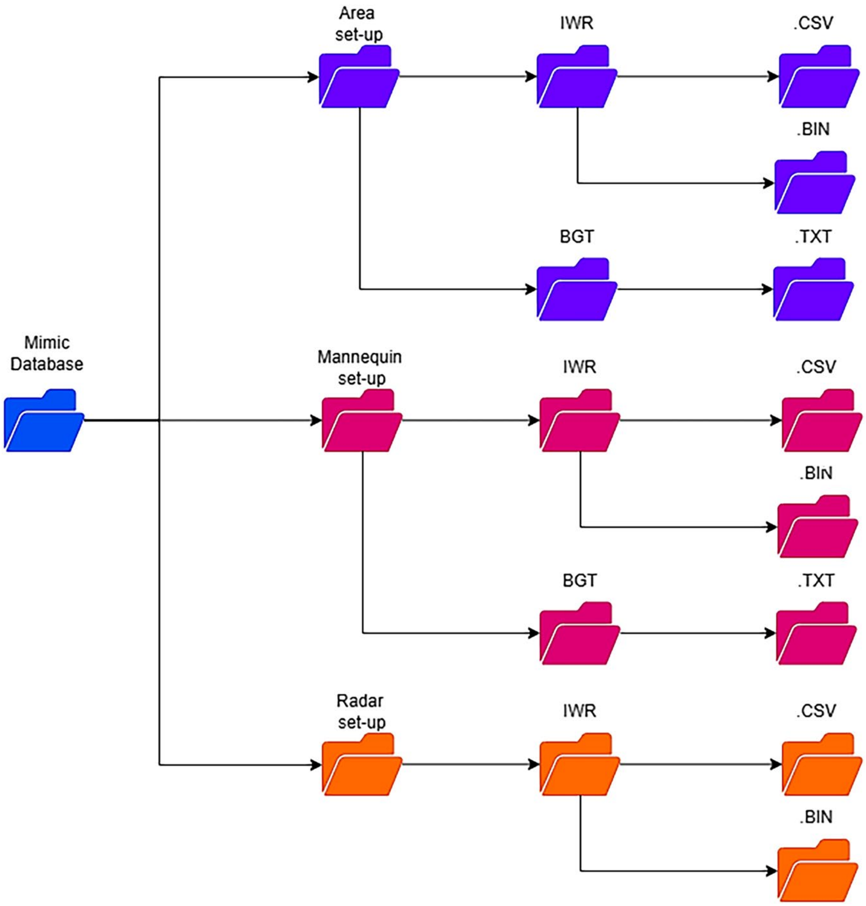

Fig. 3 Folder structure of the SiVis database.

Encode	Description	Example
`[Meas#]`	Measurement ID	022
`RadarPosition`	Radar mounting	Top
`Distance`	Height distance to targets	3m
`Orientation`	Radar angle	45°
`SimMan[#]`	Number of simulators present	SimMan2
`Cond`	Conditions assigned to each SimMan	CondA, CondB, CondC
`ADCsamples`	Number of ADC samples	ADC256
`Chirpxxx`	Chirps per loop	Chirp128
`SRxxxx`	Sample rate in ksps	SR1000
`RExx`	Ramp end time in μs	RE60
`FPxx`	Frame periodicity in ms	FP40
`Gainxx`	Receiver gain in dB	Gain30
`FSxx`	Frequency slope in MHz/μs	FS30

Table 1. Encoding of files from SiVis Dataset.

[Meas#]_RadarPosition_Distance_Orientation_SimMan[#]_ Condition_SimA[#]_ Condition_SimB[#]_ ADCsamples_Chirps PerLoop_SampleRate_RampEndTime_FramePeriodicity_RxGain_FreqSlope

The description of the encoding is shown with example values in Table 1.

Technical Validation

While no external ECG or respiratory belt was used, the simulator's programmed vital sign parameters serve as exact ground truth references, providing deterministic, reproducible benchmarking conditions without human variability. Because each mannequin's vital parameters were digitally programmed and time locked to radar recordings, the ground truth HR and RR are precisely defined, ensuring rigorous validation despite the absence of wearable sensors.

As part of the contributions of this paper, we propose a pipeline as the technical validation of the dataset, to extract multipatient vital signals, which includes two algorithms: (1) target separation; and (2) vital sign extraction.

Target Separation. The goal of the target separation stage is to isolate the individual return signals from each detected reflector (patient) in the scene so that their vital sign modulations can be analyzed independently. After detecting the approximate range angle bins corresponding to each person, we apply an adaptive beamforming algorithm (MVDR) over a small slab of range bins around each detection. This process suppresses interference from other targets and clutter by steering a spatial filter toward the desired direction, and enhances the signal-to-noise ratio (SNR) of each person's return, making it possible to observe the weak phase variations due to chest motion. The process produces one complex time series per target, which contains only the motion-induced phase variations for that individual. By separating the mixture of echoes into distinct channels, we ensure that subsequent phase-processing and spectral analysis reliably yield each patient's RR and HR without any cross-contamination of signals. The process is described in Algorithm 2:

Algorithm 1 Preprocessing & Target Separation.

1: **Input:** ADC : $\mathbf{X} \in \mathbb{C}^{N_f \times N_r \times N_c \times N_a}$, M
2: **Output:** Beamformed signals $\mathbf{B} \in \mathbb{C}^{F \times M}$
3: RP $\leftarrow$ FFT(ADC, $axis = -1$) (Range profiles)
4: RAmap $\leftarrow$ ComputeRangeAngle(RP, N_θ)
5: $\mathcal{T} \leftarrow$ DetectTargets(RAmap, M, mD, B, N_θ, R_{mean})
6: $\mathbf{B} \leftarrow$ BeamformMVDR(ADC, $\mathcal{T}$, N_θ, f_c, win)
7: **Return B**

In the first part of the algorithm, we wish to pick up to M strong peaks in the Range-Angle map $\mathbf{RA} \in \mathbb{R}^{R \times A}$ while enforcing minimum separations in range and angle. Given the speed of light c, B, and mD, we compute the range resolution $\Delta R = c/(2B)$ and the minimum range-bin offset $r_{\min} = \lceil mD/\Delta R \rceil$, then the angular resolution per bin $\Delta\theta_{\text{bin}} = 180°/N_\theta$, the physical-to-angular separation $\Delta\theta = \arcsin(mD/R_{\text{mean}})$, and finally the minimum angle-bin offset $a_{\min} = \lceil \Delta\theta/\Delta\theta_{\text{bin}} \rceil$.

We then scan the flattened map in descending power order, rejecting any candidate (r, a) for which $r < r_{\min}$ or $r > R - r_{\min}$ or there exists a previously selected (r_0, a_0) such that $|r - r_0| < r_{\min}$ and $|a - a_0| < a_{\min}$, and continue this process until we have collected M targets. In the MVDR beamforming stage, For each detected (r_m, a_m), extract the slab of range-FFT bins $[r_m - \text{win},\ r_m + \text{win}]$ from RP, reshape to form the covariance matrix $\mathbf{R}$, and build the weight

$$\mathbf{w}_m = \frac{\mathbf{R}^{-1}\mathbf{a}(\theta_m)}{\mathbf{a}(\theta_m)^H\,\mathbf{R}^{-1}\mathbf{a}(\theta_m)},$$

where $\mathbf{a}(\theta_m)$ is the steering vector at angle θ_m. Once$\mathbf{w}_m$ is computed(using snapshots from all frames/chirps), we apply it to each frame's gated data. Applying $\mathbf{w}_m^H$ to each frame yields a complex time series $y_m[f]$. The Parameters definitions can be observed in Table 2.

Following the beam estimation process, HR and RR are derived from the processed radar signals, where z the complex-valued slow-time signal for each beam, length F is input for processing. HR is obtained by filtering the instantaneous phase-derived frequency (φ) within the typical cardiac frequency range (0.8-3.0 Hz) and identifying the predominant frequency in each of the beams with the power spectral density $P_h(f)$. RR estimation involves filtering the same instantaneous frequency $\Delta\varphi$ within the respiratory frequency band (0.1-0.6 Hz), extracting the envelope of this signal and detecting respiration cycles by peak identification.

The data set was analyzed for mean heart rate and breathing rate estimations in a multi-subject scenario. Furthermore, we emphasize that this work aimed to propose and validate a publicly available dataset, rather than to propose a novel methodology. Therefore, the results presented in this work is an illustration of the feasibility of vital signs estimations in the proposed SiVis database, where no attempt has been made to improve the accuracy of estimations. Therefore, we have used a basic data processing pipeline facilitating the multi-subject identification followed by HR and RR estimations, respectively.

The data processing pipeline starts with decoding the data using Algorithm 2, which resulted in $X \epsilon \mathbb{C}^{N_f \times N_r \times N_c \times N_a}$. Subsequently, multi-subject identification using Algorithm 1. Finally, the mean HR and RR estimates were calculated from the beams, which comprise inphase and quadrature signals for each subject in representative samples from the SiVis database. The qualitative and quantitative results corresponding to each stage, that is, target identification and estimation, are presented in the following sections. Specifically, target selection was presented using a spectrogram, while the estimation results were presented using four quantitative error metrics: RMSE, MAE, ErrorSD, and ME, in beats per minute, and MAPE (%) and Pearson correlation coefficients, respectively. Following, a bland-altman analysis is also presented to investigate the level of agreements between ground truth and estimated values.

Symbol	Description
M	Number of target patients (Beamforming directions or slow-time traces)
N_θ	Number of discrete angle bins used in the spatial FFT.
mD	Minimum physical separation between two targets.
B	Chirp bandwidth for radar.
R_{mean}	Used to compute angular-bin separation threshold.
f_c	Radar center frequency
win	Range-bin window half-width used in MVDR beamforming.
$\mathbf{B}$	Beamformed slow-time data matrix, shape $F \times M$.
F	Number of slow-time frames (samples) per chirp.
f_s	Slow-time sampling rate (frames per second).

Table 2. Parameters definitions for algorithm 2.

Target Selection. The accuracy of the estimation depends on the ability to identify and segregate subjects in a multi-patient scenario, which is performed by identifying suitable research bins to identify locations using range-angle maps, followed by the MVDR beamforming algorithm. The range-angle map was created on the basis of the chirp settings for each measurement, which was then used for subject identification, followed by beamforming (Algorithm 1). Figure 4 presents the spectrogram showing the range bin corresponding to the tracking of mannequin subjects in the TopFrontal 3-patient configuration. The spectrogram shows the energy at bin range 128, at the same distance from the radar setup. Accompanied by angle and minD, we can identify the patients in the scenario in (128,64), (128,54) and (128,74).

Quantitative Analysis. As mentioned above, the HR and RR estimates were analyzed using five error metricsa: RMSE, MAE, ErrorSD, ME, and MAPE, and Pearson correlation coefficient. Table 3 presents these metrics for each vital sign. It should be mentioned that the data processing pipeline used was not optimized for accuracy enhancement. Despite this, the error metrics presented here are closer to clinically accepted limits presented in medical journals such as the American Association of Sleep Medicine[24]. For instance, the clinically accepted error limits for HR measurement is ± 5 bpm, where the reported MAE is 6.60 bpm, with a difference of 1.5 bpm. Although the error standard deviation for HR estimations is higher, which is subjected to the robustness and stability of the proposed pipelines in the near future. Furthermore, the Pearson correlation coefficient (ρ) of 0.92 also justifies the good performance of the existing pipeline.

Similarly, the reported MAE for RR estimations is 1.47 bpm, which is between the clinically accepted limits of ± 3 bpm. However, the Pearson correlation coefficient for the respiration rate is low, but positively correlated with the ground truth values, which is subject to improvement in future studies. Low correlation attributed to higher variability in respiration rate due to challenging clinical conditions simulated for SiVis database, which is consistent with various studies[25]. However, we expect to develop significant improvements with respect to the metrics reported in this work with efficient and robust estimation pipelines in future studies.

Bland-Altman Analysis. Following quantitative analysis, Bland-Altman analysis is presented to provide insight into the error distribution in the ranges of vital signs. Figure 5A,B present the Bland-Altman plot for mean HR estimates. It is important to note that one data point in the graph corresponds to multiple subjects, as the clinical conditions for the SiVis database were simulated based on the respective values of vital signs. Specifically, samples collected under one clinical condition have the same vital sign value, which resulted in a limited number of data points on the plot, but not limited samples.

From Fig. 5A, the mean bias for HR estimations is -0.63 bpm, while the upper and lower statistical limits defined by $\mu \pm \sigma$ are 14.85 and -16.11 bpm, respectively. The higher statistical limits correspond to the higher error standard deviation, as reported in Table 3. Higher statistical limits were due to the complex clinical conditions under which the data was collected, which requires sophisticated data processing pipelines, unlike this work. Furthermore, the Bland-Altman test also shows that the method used is not biased towards a particular HR value or its particular ranges, despite the inclusion of abnormal HR ranges in the SiVis database.

The mean bias for the estimation of the RR is 0.54 bpm (Fig. 5B), where the upper and lower statistical limits are 3.91 and -2.84 bpm, respectively. One key observation from this plot is that the statistical limits are closer to the clinically acceptable limits, despite the fact that no attempt was made to improve the accuracy. Additionally, the scattered data points in the plot point to a limitation of the used data process pipeline, that a small number of estimated values achieved the same value as ground truth, which is expected, considering the clinical conditions covered by SiVis database. Furthermore, the prevalence of outliers with respect to the statistical limits is due to the inclusion of complex limit conditions such as asthma attacks, cardiac arrests, etc.

Usage Notes

Algorithm 2 outlines the procedure for decoding the binary file generated by mmWave Studio. We recommend using this algorithm for processing the data presented in this study.

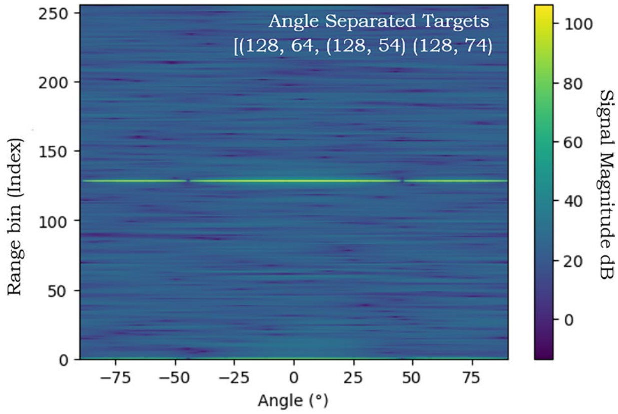

Fig. 4 Range-Angle intensity map (in dB) for a three-patient configuration showing clear separation of targets at similar range (bin ≈ 128) but distinct angles. Detected target coordinates [(128, 64), (128, 54), (128, 74)] correspond to individual mannequins. The colorbar indicates signal magnitude in decibels (dB).

Vital Signs	RMSE	MAE	ErrorSD	ME	MAPE (%)	ρ
HR	7.91	6.60	7.89	−0.63	6.11	0.92
RR	1.80	1.47	1.71	0.54	8.43	0.72

Table 3. Summary of Error metrics on SiVis database.

Example applications include testing of vital sign separation algorithms (e.g., ICA, MVDR, MUSIC), evaluating robustness across radar configurations, and training machine learning models for multi-patient tracking[26].

Algorithm 2 Decoding IWR6843 Raw Data (non Interleaved Complex Format).

Input: binary file path P; receivers N_r; chirps per frame N_c; ADC samples per chirp N_a
Output: $\mathbf{X} \in \mathbb{C}^{N_f \times N_r \times N_c \times N_a}$
Read binary file P as int16 array $\mathbf{d}$
Reshape $\mathbf{d}$ to $[-1, 2]$ and form complex samples: $\mathbf{c} \leftarrow \mathbf{d}[:, 0] + j\,\mathbf{d}[:, 1]$
$S \leftarrow N_r \times N_c \times N_a$
$N_f \leftarrow \mathrm{len}(\mathbf{c})/S$
Initialize $\mathbf{X}$ as complex array of shape $N_f \times N_r \times N_c \times N_a$
for $f = 0$ **to** $N_f - 1$ **do**
  for $r = 0$ **to** $N_r - 1$ **do**
    $s \leftarrow fS + r\,N_c N_a$
    $e \leftarrow s + N_c N_a$
    $\mathbf{y} \leftarrow \mathbf{c}[s : e]$
    Reshape $\mathbf{y}$ to $[N_c, N_a]$
    $\mathbf{X}[f, r, :, :] \leftarrow \mathbf{y}$
  end for
end for
return X

The data set can be used for developing signal processing methods two different perspectives: radar configurations and complex clinical conditions. SiVis integrates extensive variability across radar parameters (e.g., chirp configurations, sampling rates, gain settings), spatial arrangements (e.g., top, frontal, and lateral views at multiple distances), and clinical subject conditions, respectively. Any processing pipeline should adjust its parameters according to the data recording conditions.

The preliminary analysis resulting from the basic data processing pipeline in the SiVis database had parameters adjustment according to this recording conditions, demonstrating the feasibility of HR and RR estimations in the SiVis database, with metrics closer to clinically accepted thresholds.

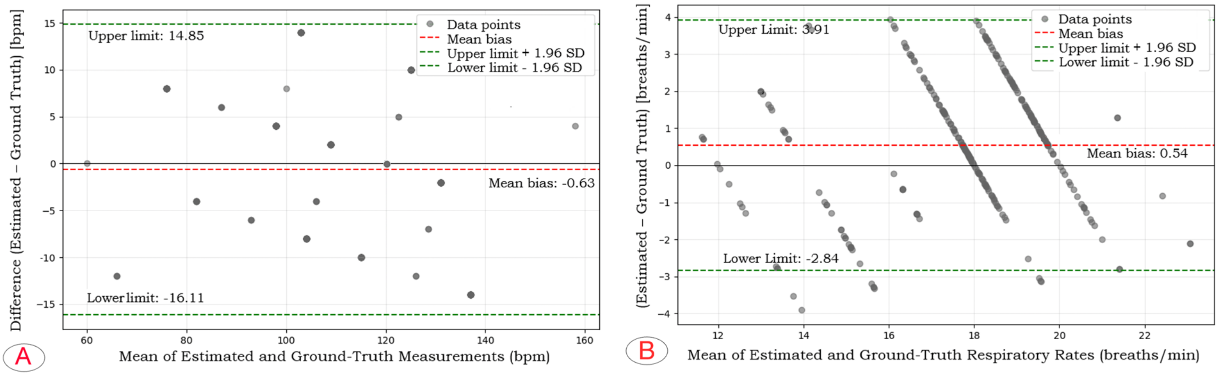

Fig. 5 Comparison of Bland-Altman plots for HR and RR estimations on the SiVis database. (**A**) Bland-Altman plot comparing radar-estimated and mannequin-programmed heart rates. The red dashed line shows the mean bias (0.63 bpm), and green dashed lines denote the ± 1.96 SD limits (upper = 14.85 bpm, lower = 16.11 bpm). Each point represents an averaged estimate per clinical condition. The majority of values lie within clinically acceptable limits. (**B**) Bland-Altman plot comparing radar-estimated and mannequin-programmed respiratory rates. The red dashed line shows the mean bias (0.54 bpm), while green dashed lines represent the ± 1.96 SD limits (upper = 3.91 bpm, lower = 2.84 bpm). Most values remain within the clinically acceptable range (± 3 bpm).

Data availability

The dataset is deposited in the Harvard Dataverse repository as SiViS - Simulated multi-patient physiological clinical states[22], available at https://doi.org/10.7910/DVN/S9SNAK.

Code availability

The decoding the data code is available in the main files of the dataset, a Python script, provided in JSON format. This script enables decoding and processing of the raw data collected during the experiments, facilitating further analysis.

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Author contributions

K.M.R.L., M.C. and P.M. design the study. K.M.R.L. and A. GR conceived the dataset,. K.M.R.L., A.G. and M.C. developed the decoding pipeline and wrote the paper; S.P., J.Š. and I.Z. designed the clinical approach and validated the data; all authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

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